

Summary

- Software Engineer with around 7+ years of experience in object-oriented design and web development across finance, healthcare, and ecommerce industries. Proficient in Frontend and Backend development, Database administration, and Automation testing. Able to create scalable online solutions and enhance user engagement strategically.
- Proficient in Cloud-based deployment on Amazon EC2 and their other offerings like S3 and AWS including EC2, VPC, S3, EBS, ELB, Cloud-Front, IAM, RDS and CloudWatch, Elastic cache, SNS, SQS, AMI.
- Cloud Engineer with extensive experience designing, deploying, and managing scalable cloud-native applications using AWS services. Proficient in backend development with Java, microservices architecture, and DevOps practices including CI/CD, containerization, and Infrastructure as Code. Adept at integrating Spring-based applications with AWS services to deliver secure, high-performing solutions
- Proficient in cloud-based deployment on Microsoft Azure and its various services including Azure Virtual Machines (VMs), Virtual Network (VNet), Blob Storage, Azure Load Balancer, Azure Active Directory (AAD), Azure SQL Database, Azure Monitor, Azure Functions, App Services, Key Vault, Event Grid, Service Bus.
- Experience in Web based UI's using Angular, JS, HTML, HTML5, CSS, CSS3, Bootstrap, JavaScript.
- Hands-on experience in Spring Frameworks, including Spring 5, Spring MVC, Spring Data, Spring Boot, Spring Security, Spring Cloud.
- Hands-on experience in creation of REST Microservice APIs using Spring Boot Application.
- Proficient in MVC, Singleton, Session Facade, Service Locator, Visitor, Observer, Decorator, DAO design patterns.
- Well experienced in Core Java – asynchronous programming, multithreading, collections and a few design patterns.
- Used Core java8 and java11 in development of enterprise applications.
- Experience in delivery of prioritized solutions and customer support via NodeJS microservice architecture which includes modules for call routing, chat boxing, analytics, and scripted messaging using Docker Containerization.
- Experienced with RDBMS implementation and development using Oracle, My SQL, MSSQL and DB2.
- Extensively used Java8 features such as lambda expressions, method referencing, Stream API's, parallel operations on collections, multithreading, and effective sorting mechanisms and to increase the performance of the application.
- Strong experience in Spring Framework modules such as Spring MVC, Spring JDBC, Spring Boot, Spring Microservices, Spring Dashboard, Spring Eureka, Spring Netflix, and Spring Batch.
- Experienced in frameworks like Flask, Django and Python packages like PySide, PyQtGraph, NumPy, MatPlot Lib
- Designed and optimized complex SQL queries and stored procedures to improve application performance and data integrity in PostgreSQL and Oracle DB environments.
- Experience in setting up end to end logging, monitoring and alerting configurations for RESTful APIs to track their performance
- Expert in working with relational databases such as Oracle, MySQL, DB2, Aurora PostgreSQL, and SQL Server to generate complex SQL and PL/SQL queries, including joins, subqueries, views, and triggers.
- Hands-on experience in unit testing using JUnit, along with mocking frameworks like Mockito and Power Mockito for behavior-driven development.
- Designed, implemented, and maintained end-to-end CI/CD pipelines using tools such as Jenkins, Team Forge, and TFS, significantly streamlining and automating the software release process.
- Proficient with deployment and containerization tools, including CI/CD workflows, Docker images, Docker containers, and Kubernetes for orchestrating scalable microservices.
- Extensive experience with version control systems, particularly Git, and proficient in using source code management tools like GitHub and Bitbucket for collaborative development and code review.

PROFESSIONAL EXPERIENCE

AT&T

Software Engineer

April 2024 – Present

Plano, Texas

- Generating various capacity planning reports (graphical) using Python packages like Numpy, matplotlib.
- Analyzing various logs that are been generating and predicting/for forecasting next occurrence of events with various Python libraries.
- Designed and deployed a microservices architecture using Flask for document ingestion, Fast API for real-time inference, containerized via Docker, and orchestrated on Kubernetes with Helm for scalable, resilient operations.

- Integrated multiple data layers by combining relational databases (PostgreSQL, SQL Server) with NoSQL stores (MongoDB, Cassandra) and in-memory caches (Redis) for an efficient, multi-model persistence strategy.
- Engineered asynchronous data processing workflows utilizing Celery, RabbitMQ, and Python's asyncio to manage high-volume tasks such as embedding generation, data preprocessing, and distributed logging.
- Implemented semantic search pipelines that process textual data with NLP libraries (NLTK, spaCy), generate embeddings via the OpenAI API, and perform similarity matching using vector databases.
- Enhanced observability with advanced logging and monitoring tools such as Prometheus, Grafana, and the ELK stack (Elasticsearch, Logstash, Kibana) to track system performance and diagnose issues in real time.
- Secured microservices endpoints by implementing JWT- and OAuth2-based authentication and authorization, enforcing HTTPS/TLS encryption and API gateway protections for data safety
- Developed comprehensive RESTful and GraphQL APIs with FastAPI and Flask, offering versioned endpoints, dynamic response generation, and self-documenting Swagger/Open API interfaces.
- Engineered asynchronous data processing workflows utilizing Celery, RabbitMQ, and Python's asyncio to manage high-volume tasks such as embedding generation, data preprocessing, and distributed logging.
- Implemented streaming data pipelines with Apache Kafka and Spark, facilitating real-time analytics on user interactions, chatbot engagement, and feedback-driven iterative improvements.
- Utilized distributed caching mechanisms (Redis and Memcached) to store frequently accessed data, reducing database load and minimizing response latencies during high-traffic operations.
- Synchronized multiple data ecosystems by unifying SQL (PostgreSQL, Aurora) and NoSQL (MongoDB, Cassandra) repositories, enabling consistent query performance and cross-platform data integrity.
- Streamlined dependency management by transitioning from Java's Maven/Gradle to Python's pip and Poetry, fostering a lightweight and agile development environment.
- Redesigned database connectivity layers by shifting from Java JDBC configurations to Python-based ORMs (SQLAlchemy), optimizing SQL query performance across PostgreSQL and MySQL
- Enhanced logging and monitoring using Python's logging libraries, integrated with Azure Monitor and the ELK stack, to match and exceed the visibility of prior Java implementations.
- Enabled fine-grained role-based access control (RBAC) across microservices and Kubernetes clusters to enforce secure, permission-based interactions among distributed components.
- Scaled processing dynamically using Azure's serverless compute environment, which automatically adjusts resources for peak loads during embedding generation and model fine-tuning.
- Optimized container orchestration on AKS, using native monitoring and logging tools to enhance system reliability and resource utilization.

Environment: Java, Python, JavaScript, TypeScript, SQL, HTML, CSS, Bash, Shell Scriptin, Spring Boot, Spring MVC, Spring Data, Spring Security, Spring Cloud, Spring Batch, Flask, FastAPI, Azure (VMs, Blob Storage, Azure Functions, Azure SQL Database, ARM Templates, Virtual Network, Azure DNS, Azure Monitor, Azure Active Directory, Event Grid, Cosmos DB), Microsoft Azure, JUnit, Mockito, Power Mockito, Test-Driven Development (TDD), Behavior-Driven Development (BDD), Selenium

Amazon

July 2022 – Feb 2024

Software Development Engineer

Arlington, Virginia

- Designed and deployed microservices-based applications using Java and Spring Boot on AWS using EC2, Elastic Beanstalk, and Lambda.
- Developed secure authentication and authorization using Spring Security integrated with AWS Cognito.
- Automated infrastructure provisioning with AWS CDK and CloudFormation, following Infrastructure as Code best practices.
- Built and managed CI/CD pipelines using CodePipeline, Code Build, and CodeDeploy for Java-based applications.
- Integrated applications with RDS and DynamoDB using Spring Data.
- Enabled high availability and scalability using Auto Scaling Groups and Elastic Load Balancer within a well-architected VPC.
- Containerized applications with Docker and deployed via ECS (Fargate) and EKS clusters.
- Used CloudWatch, X-Ray, and CloudTrail for performance monitoring, tracing, and auditing.
- Leveraged AWS Secrets Manager and Systems Manager Parameter Store for secure configuration management.
- Managed asynchronous workflows and event-driven architectures with SQS, SNS, and Event Bridge.
- Implemented core features of AngularJS framework such as dependency Injection, Databinding, Filters, Directives, Templates, Services and Deep Linking.
- Created AngularJS controllers, services and used AngularJS filters for filter functionality in search box and integrator functionalities.

- Architected a Redis caching system with OAuth2 token-based authentication using Spring Security to improve efficiency of the RESTful API for Kotlin mobile apps; achieved a 50% decrease in data retrieval latency, enhancing user experience and scalability.
- Improved overall performance via multithreading, collections, synchronization, and exception handling strategies for complex data computations and analysis.
- Developed REST architecture-based web services using JAX-WS to facilitate communication between client and servers.
- Involved in development of web services for business operations using SOAP, WSDL, JAX-WS, JDOM, XML, XSL and Apache CXF.
- container orchestration platforms by deploying services on Kubernetes clusters (EKS/GKE), using Helm charts and Docker Swarm for automated scaling, load balancing, and zero-downtime updates.
- Designed and deployed robust, asynchronous RESTful APIs using FastAPI (with Pydantic validation) and Flask, ensuring efficient communication between chatbot interfaces, core logic services, and various data stores within a Docker/Kubernetes containerized environment.
- Implemented various design patterns: Singleton, Data Access Object (DAO), Command Design Pattern, Abstract Factory, Factory Method Design Pattern
- Experience in handling Tomcat with Apache Web Server, BEA Web Logic, IBM Web Sphere and JBoss Application Servers.
- Proficient in RDBMS implementation using Oracle, MySQL, MSSQL, and DB2, as well as in NoSQL database modeling and development with MongoDB and Cassandra.
- Expert in designing and integrating robust persistence layers with frameworks like JPA and Hibernate.
- Skilled in using version control and configuration management tools including GIT, SVN, and CVS while automating builds with tools such as Ant and Maven.
- Experienced in containerizing and deploying microservices via Docker, with end-to-end CI/CD pipeline setups utilizing Jenkins and GitHub on AWS or other cloud platforms.
- Involved in setting up the end-to-end architecture for new APIs – onboarding them to API gateway, routing using Route53, Load Balancing, configurations to deploy to AWS Lambda and failover strategies.
- Implemented a library used to persist user session data using AWS DynamoDB.
- Deployed several APIs to AWS Lambda and setup monitoring and alerting for the same.
- Developed a library to capture various metrics of APIs for applications like Number of Requests, Number of Responses, etc.
- Implemented dashboards for several APIs using AWS CloudWatch and New Relic that are used to monitor the application performance.
- Implemented scaling, monitoring and alerting for several APIs and cloud components to ensure APIs scale on increased traffic and alert when something goes wrong immediately.
- Involved in implementation of CI/CD pipeline to deploy vendor applications and orchestrator APIs on cloud which improved operational efficiency
- Extensive hands-on expertise in Amazon Web Services, working with EC2, S3, Elastic Beanstalk, Lambda, RDS, VPC, Direct Connect, Route53, CloudWatch, CloudFormation, IAM, SNS, and DynamoDB, plus integrating AWS solutions within broader application infrastructures.
- Competent in implementing robust messaging systems using Kafka and Kafka event sourcing to facilitate reliable and scalable communication.
- Established responsive user interfaces using AngularJS framework & equipped advanced features like lazy loading, routing, and form validation.
- Invoked GraphQL in frontend with Node.js to handle data within AngularJS applications and engaged TypeScript for type safety & mirroring skills in web development. Employed Grafana and Prometheus as visualization tools for real-time monitoring, ensuring system health.
- Collaborated cross-functionality to translate design mock-ups into interactive user interfaces (UI), implementing HTML5, CSS3, JavaScript (JS), XML & jQuery to enhance user satisfaction scores by 80%.
- Enhanced data management system utilizing Hibernate & Spring Framework, integrating MongoDB with Sharding to scale CRUD operations and complex query efficiency also containerized services by deploying Docker images through Azure Kubernetes clusters with Azure Container Service (ACS).

Environment: Java, Python, JavaScript, TypeScript, SQL, HTML, CSS, Bash, Shell Scriptin, Spring Boot, Spring MVC, Spring Data, Spring Security, Spring Cloud, Spring Batch, Flask, FastAPI, AWS (EC2, S3, Lambda, RDS, CloudFormation, VPC, Route 53, CloudWatch, IAM, SNS, DynamoDB), Microsoft Azure, JUnit, Mockito, Power Mockito, Test-Driven Development (TDD), Behavior-Driven Development (BDD), Selenium

- Directed Agile software development operations through Scrum practices and devised tasks in Jira; created Linux bash scripts and Python automation scripts for swift deployment, log analysis, and system monitoring, resulting in a substantial reduction in project completion time.
- Field security measures across system architecture through the successful deployment of SQL databases and SSL certificates via OpenSSL, bolstering VPC networks to Kafka clients and enhancing overall data protection.
- Architected Test-Driven Development strategies, configuring EC2 servers with ELB storage, Nginx, Fail2Ban & File beat. Employed Java frameworks including Spring MVC & Maven to optimize system performance through unit testing, ensuring code integrity & scalability.
- Orchestrated asynchronous API calls and parallel processing by leveraging Python's asyncio and Fast API's native asynchronous support, ensuring high responsiveness for concurrent user requests.
- Migrated core microservices from Java (Spring Boot) to Python, leveraging frameworks like Fast API and Flask to streamline code complexity and accelerate feature development
- Integrated asynchronous message queues via RabbitMQ, enabling Python microservices to decouple processes originally managed by synchronous Java messaging patterns.
- Skilled in front-end web-based applications, mobile apps such as cross-browser optimization, performance optimization (including for mobile devices on iOS and Android), web accessibility, search engine optimization (SEO).
- Maintained and optimized both Angular.js and React.js codebases for web development. Created responsive UIs with HTML, CSS, JavaScript, jQuery, TypeScript for AngularJS for React while utilizing Figma/Blender animation tools.
- Originated a scalable Spring Boot application with an Oracle database (SQL), featuring customizable APIs and combined schedulers to generate comprehensive system availability reports at specific times, resulting in a 35% increase in system performance.
- Upgraded system health check APIs with AWS Route53, resulting in a 15% reduction in downtime by monitoring instances and ELB status. Configured AWS SNS notifications for immediate alerts, reducing response time by 40%.
- Created reusable React components and managed application state with Redux.
- Applied redux middleware using redux-promise, redux-thunk and redux-saga.
- Involved in development of project back-end logic layer by using most of the core java features such as Collection Framework, Interfaces and Exception Handling programming and Java 8 features.
- Built Maven application using spring boot. Used Jenkins in cross platforms and deployed EAR to WAS servers in related environments.
- Developed Data access layer using Spring Data and Hibernate to access and update information in the database.
- Experience in JSON to XML transformations and vice versa to support different services of application and involved to test REST APIs using Postman and SOAPUI.
- Used Lombok Java library, annotations to refactor and build efficient quality code by avoiding getters and setter's implementation.
- Use of tools such as PMD, find bugs, Jenkins, Sonar to ensure the continuous integration in the project.
- Developed Unit tests for applications using Mockito.
- Experience in working with various spring framework modules like Spring Core, Spring IOC, Spring Data JPA, Spring AOP, Spring Security, Spring Transaction, Spring JDBC, Spring MVC, Spring Integration.
- Developed batch jobs using Spring Batch and Develop email notification system for system failures using Spring Mail.
- Supported bug-fixing, enhancements, troubleshooting various application errors, exceptions and roadblocks on different environments and implementation of new requirements.
- Experience in using Hibernate and Spring Framework modules like Spring Core container for Dependency Injection, Application context, Spring ORM and POJO objects.
- Implemented application-level core API's as Restful web services and developed test cases using Spring Mockito for the developed web services.
- Developed messaging engine using Spring JMS and Spring AOP for audit logging.

Environment: Java, Python, JavaScript, TypeScript, SQL, HTML, CSS, Bash, Shell Scriptin, Spring Boot, Spring MVC, Spring Data, Spring Security, Spring Cloud, Spring Batch, Flask, Fast API, AWS (EC2, S3, Lambda, RDS, CloudFormation, VPC, Route 53, CloudWatch, IAM, SNS, DynamoDB), Microsoft Azure, JUnit, Mockito, Power Mockito, Test-Driven Development (TDD)

Cloud Technologies

Associate Software Engineer

December 2017– January 2021

Hyderabad, India

- Collaborated with a team to develop a web application using Java and Spring framework, adhering to Agile development methodologies.
- Proficient in Java with expertise in Multithreading, Concurrency, Collections Framework and developed secure RESTful APIs employing token-based authentication through Spring Security to safeguard sensitive customer data.
- Proposed Java and Spring annotations for backend processes, refining web communication, business logic, database interactions achieved a reduction in system errors and increased in overall performance and user experience.

- Employed Hibernate and JPA frameworks to optimize SQL database queries, improving in query execution time& integrated AWS cloud services to enhance scalability, reliability, efficiency and engineered interactive user interfaces with HTML, CSS3, JavaScript, and Angular, increasing user engagement by 40%.
- Worked on implementation of MVC pattern using XML configurations and annotations. Used controllers to handle requests and Model to send values to user interface.
- Involved in developing JSP pages for the web tier and validating the client data using JavaScript.
- Proficient in Core Java concepts like Collections, Multithreading, Serialization, and Java Beans and Streams.
- Developed single page applications using Angular, Typescript, HTML5, CSS3, and Bootstrap.
- Involved in designing, developing and testing web applications by using the HTML5, CSS3,
- Project migration into cloud environment using AWS, Docker container.
- Installation and configuration of Kubernetes, clustering them and managed local deployments in Kubernetes. Developed RESTful Services with JAX-RS using Spring Boot and Microservices Architecture.
- Developed Java API to connect with AWS S3 Services for storing and retrieving data stored in the S3 buckets.
- Developed and maintained the Continuous Integration and deployment systems using Jenkins, ANT, Maven.
- Deployed applications into Continuous Integration environments like Jenkins to integrate and deploy code on CI environments for development testing. Design and develop Test cases for TDD and BDD.

Environment: Spring Framework, Spring Boot, Spring Security, Spring MVC, Hibernate, JPA, Angular, Bootstrap, JAX-RS, S3, EC2, Lambda, Docker, Kubernetes, HTML5, CSS3, JavaScript, JSP, Angular, Bootstrap, RESTful APIs, JAX-RS, Docker, Kubernetes, JIRA, SQL Query Optimization, UI/UX

Technical Skills

Programming Languages	Java, Python, JavaScript, TypeScript, SQL, HTML, CSS, Bash, Shell Scripting
Web Development	Angular, React.js, HTML5, CSS3, Bootstrap, JavaScript, jQuery, AJAX, RESTful APIs, GraphQL, Node.js
Backend Frameworks	Spring Boot, Spring MVC, Spring Data, Spring Security, Spring Cloud, Spring Batch, Flask, FastAPI
Cloud Platforms	AWS (EC2, S3, Lambda, RDS, CloudFormation, VPC, Route 53, CloudWatch, IAM, SNS, DynamoDB), Microsoft Azure
Database Technologies	Oracle, PostgreSQL, MySQL, MSSQL, DB2, MongoDB, Cassandra, Redis, Aurora PostgreSQL
Containerization & Orchestration	Docker, Kubernetes, Helm, Docker Swarm, Azure Kubernetes Service (AKS), AWS EKS
DevOps & CI/CD	Jenkins, Maven, Git, GitHub, Bitbucket, Docker, Kubernetes, Terraform, AWS CodePipeline, SonarQube
Microservices	RESTful APIs, Microservices Architecture, Spring Boot, FastAPI, Flask, GraphQL
Automation & Testing	JUnit, Mockito, Power Mockito, Test-Driven Development (TDD), Behavior-Driven Development (BDD), Selenium
Cloud Deployment & Infrastructure	AWS (EC2, Lambda, S3, VPC, CloudWatch, CloudFormation), Microsoft Azure (VMs, App Services, Azure SQL DB)

Education

Southeast Missouri State University
Master of Science, Applied Computer Science

January 2021 – May 2022
Cape Girardeau, Missouri

Jawaharlal Nehru Technological University (JNTU)
Bachelor of Science, Computer Science

June 2014 – May 2018
Hyderabad, India